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Roll No.

Degree: RTECH - ECAM

Semester: 3rd

END-SEMESTER THEORY EXAMINATION, NOV-DEC - 2025

Course Code: EAETC305

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 03 Hours

Max. Marks: 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	i. If $x = 29.25$ and $y = 15.5$, find their binary sum and difference using 2's complement method. ii. Find BCD sum of 462 and 537	3 2	CO1
1b	Minimize using Boolean Algebra $F = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C.D + \bar{A}B.C.D + A.\bar{B}.\bar{C}.D + A.B.\bar{C}.D + A.B.C.D$	5	CO1
1c	Design a binary comparator circuit with inputs (A_1A_0, B_1B_0) & outputs ($A > B, A = B$ and $A < B$)	5	CO1,2
Q 2	Attempt any 2 parts of the following		
2a	Minimize the function using K-map, and implement using NAND-NAND logic: $F(P, Q, R, S) = \sum \min(0, 2, 8, 12) + d(4, 9, 10, 11)$	5	CO2
2b	Implement a full adder using a 1:8 De-Multiplexer.	5	CO2
2c	Implement using 4:1 MUX $F(A, B, C, D) = \sum \min(0, 2, 3, 6, 8, 12, 13, 15)$.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Design a Mod-12 asynchronous counter and draw its waveforms.	5	CO3
3b	Draw and explain 3-bit bi-directional SISO shift register.	5	CO3
3c	Design a synchronous 3-bit up counter, using J-K Flip-flops; and draw its waveforms.	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	Explain the importance of entity-architecture pair in VHDL.	5	CO4
4b	Write the VHDL Program for 8 to 1 Multiplexer using case statement.	5	CO4

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4c	Write the VHDL module code for JK flip-flop.	5	CO4
Q 5	Attempt any 2 parts of the following		
5a	Differentiate Mealy and Moore FSM with example and Draw the Mealy state diagram to detect the sequence 101.	2 + 3	CO5
5b	Draw the CMOS circuit diagram to implement the given function $F = \overline{(AB + (C + D))}$	5	CO5
5c	Write notes on the following: 1. Noise margin and Fan-out 2. Programmable logic devices.	5	CO5

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